Insight Mathematics

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for confidence
for creativity
for insight

Circular functions 1

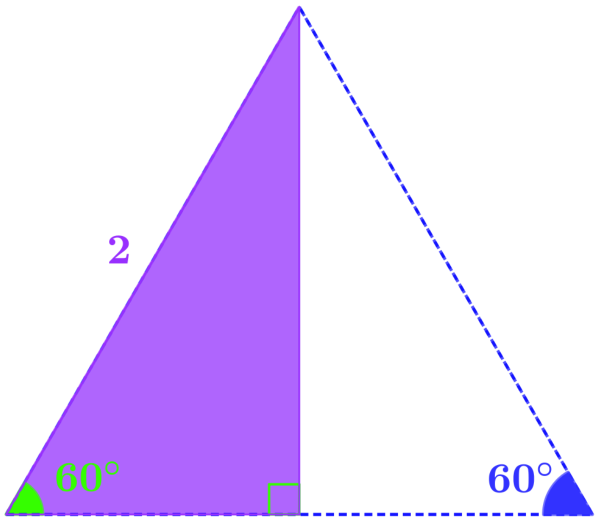
Defining the circular functions

sin, cos, tan and the unit circle

student version

First, let's think about the name for this topic. It's usually called "trigonometry". However, "trigon" is Greek for a three-sided shape, and "metry" is to do with measurement, so "trigonometry" means "measurement of triangles". Here, however, we are trying to decouple the functions from the triangles, so the name "circular functions" makes far more sense.

Use this diagram to find sin, cos, and tan of 60° and 30°.



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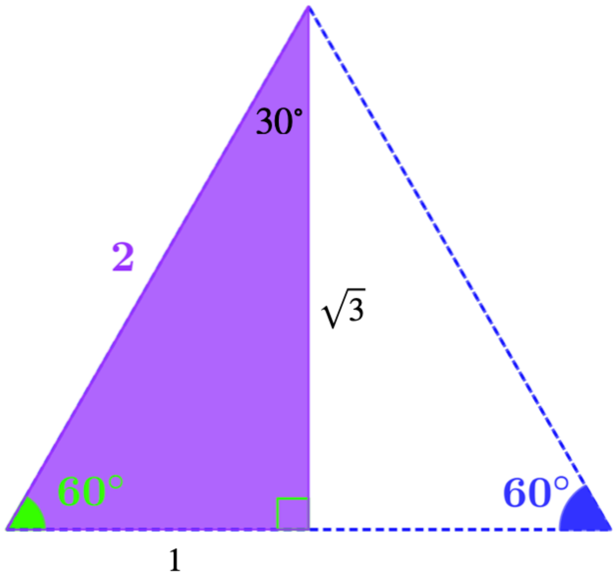
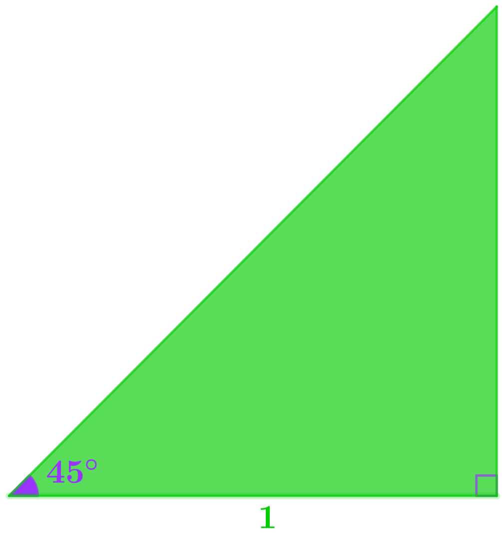
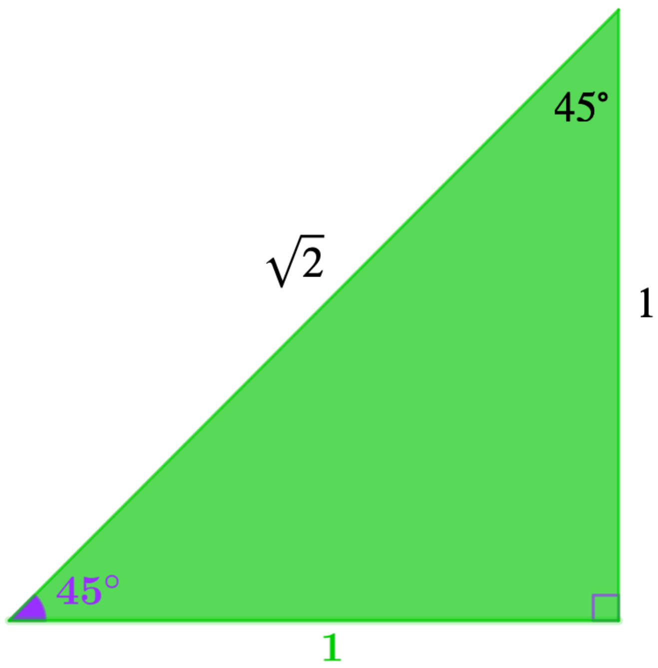


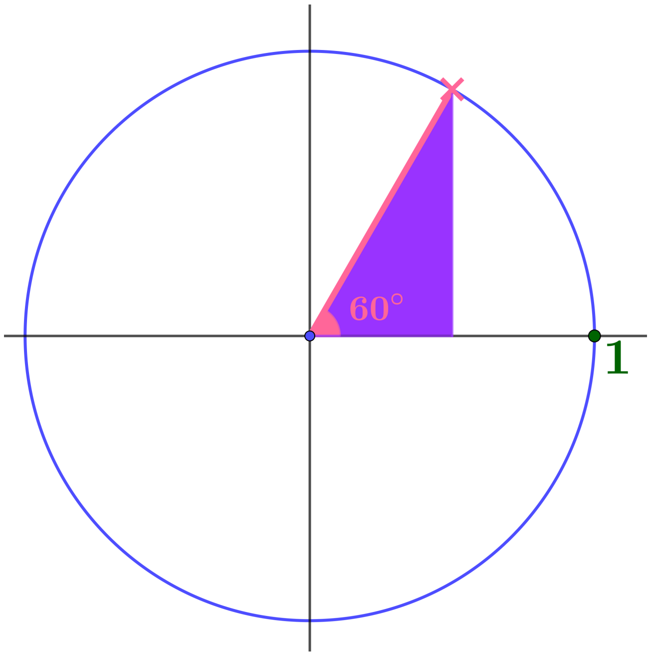
Use this diagram to find sin, cos, and tan of 45°.



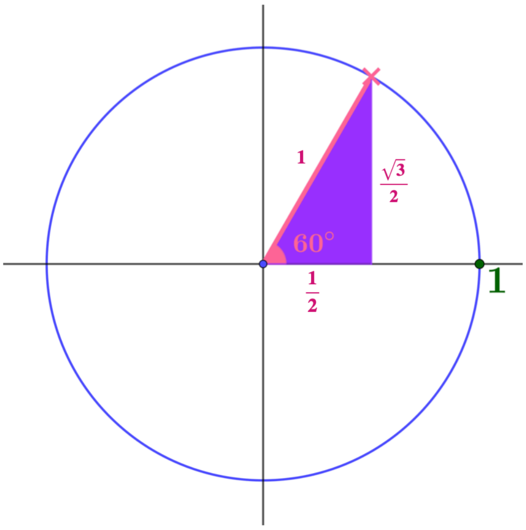
Use this diagram to find sin, cos, and tan of 45°.

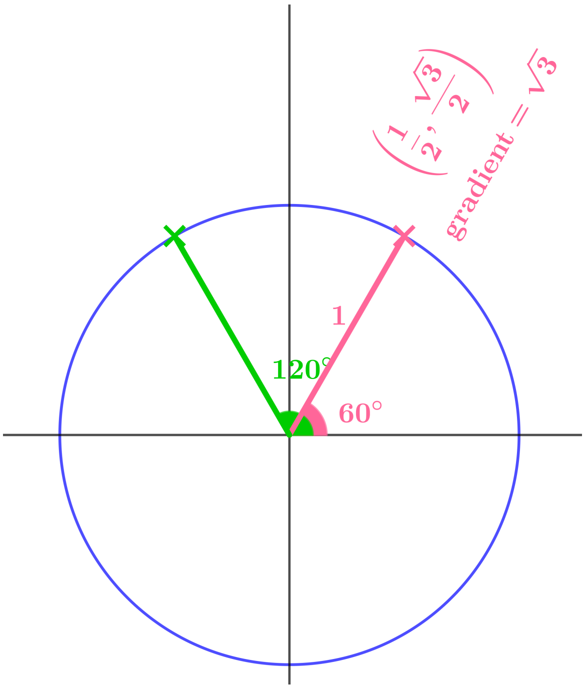


What are the coordinates of the pink point and the gradient of the pink radius?



What are the coordinates of the pink point and the gradient of the pink radius?



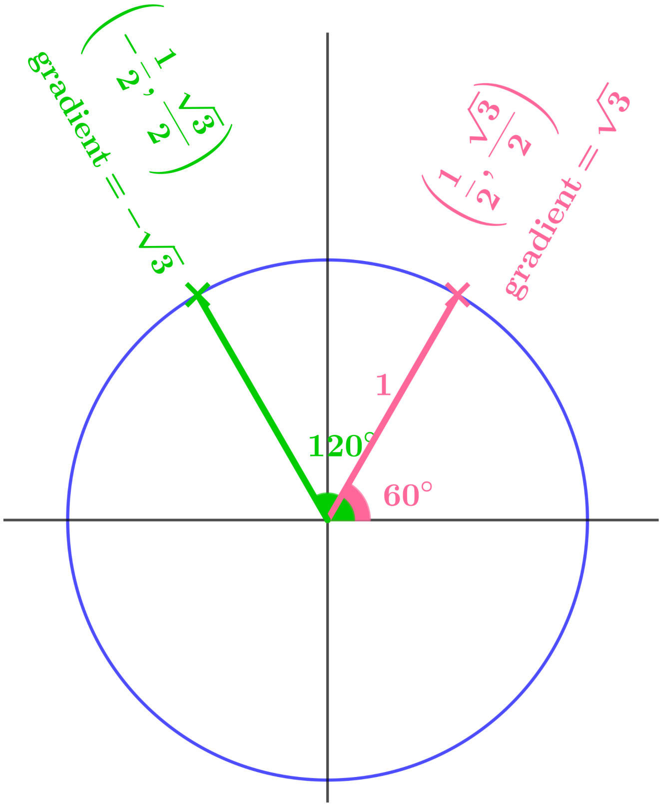


What are the coordinates of the green point and the gradient of the green radius?

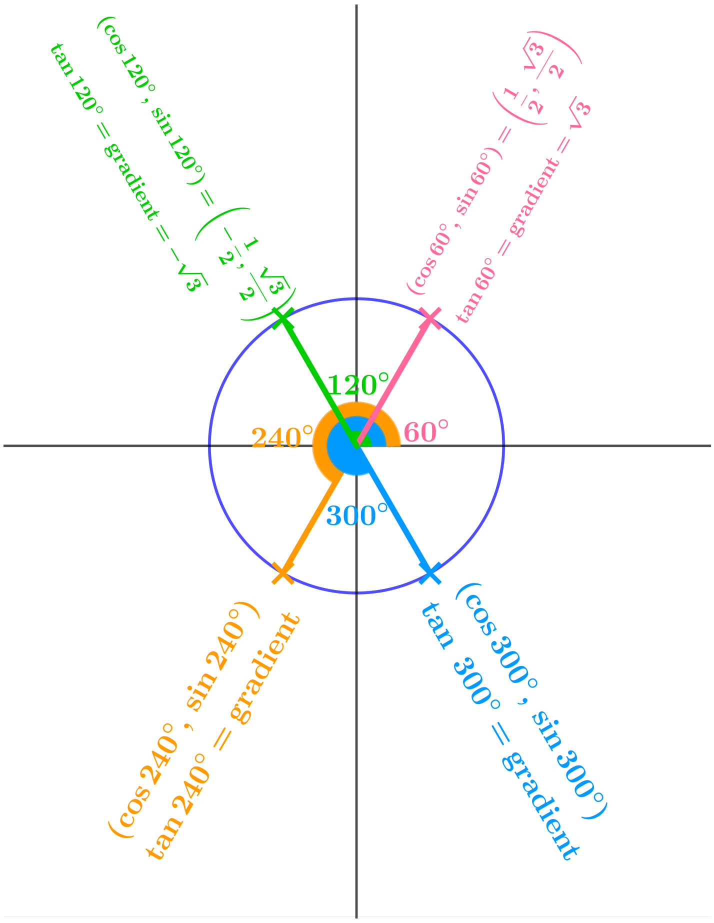
Suggest values for sin, cos, and tan of 120°.

What are the coordinates of the green point and the gradient of the green radius?

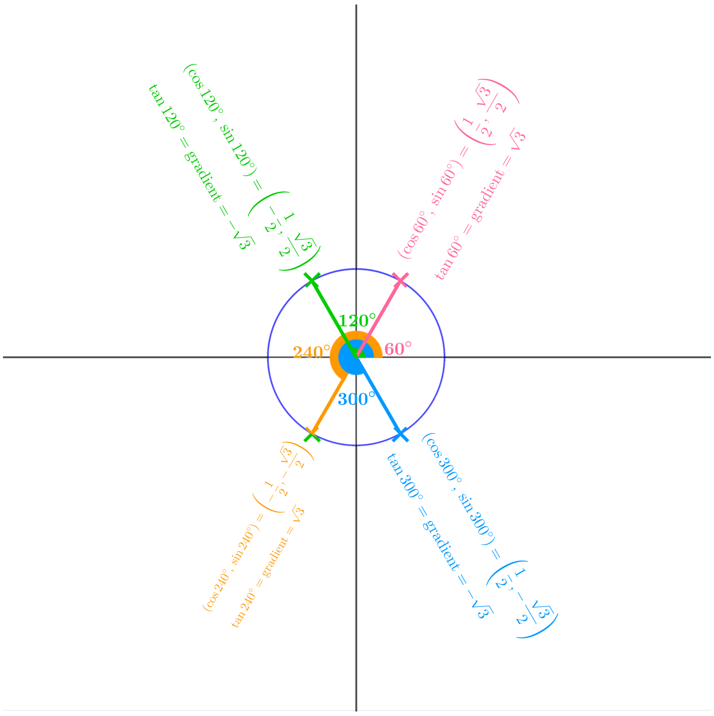
Suggest values for sin, cos, and tan of 120°.



Suggest values for sin, cos, and tan of 240° and 300°.

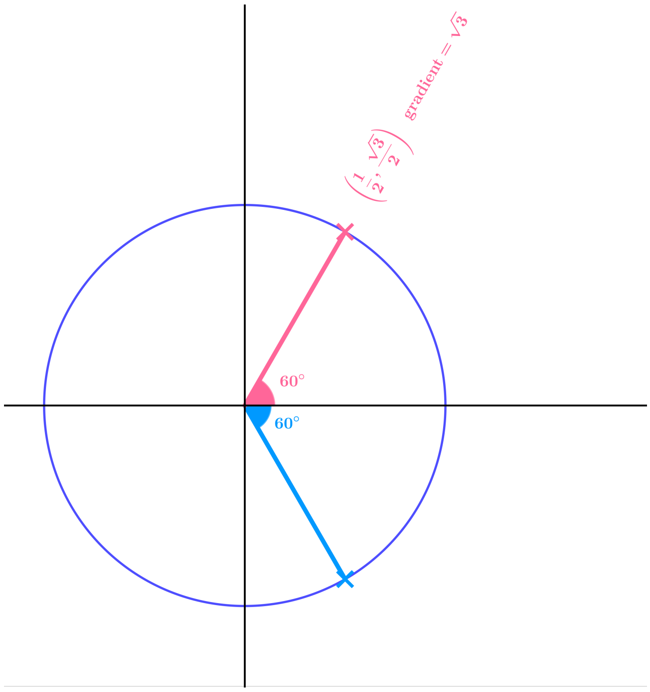


Suggest values for sin, cos, and tan of 240° and 300°.

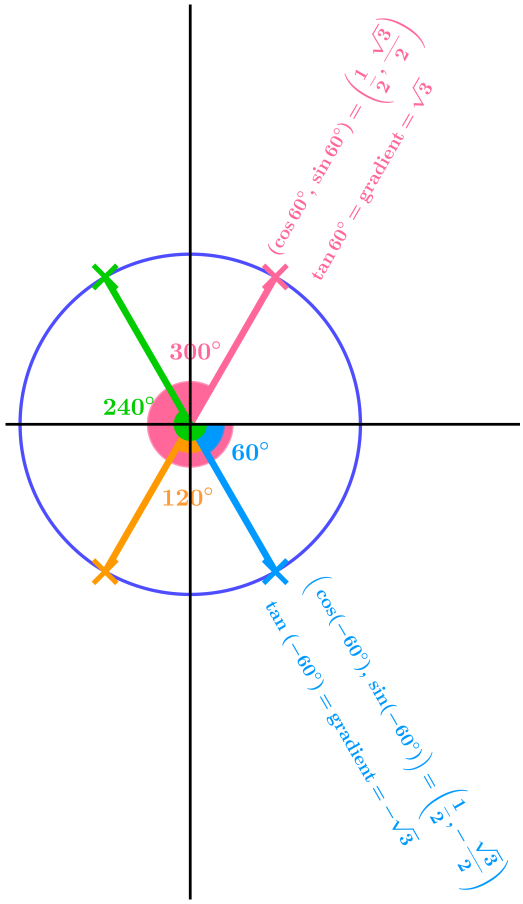


What are sin, cos, tan of $420°$, $780°$, $1140°$, $1500°$?

Suggest values for sin, cos, tan of $-60°$.

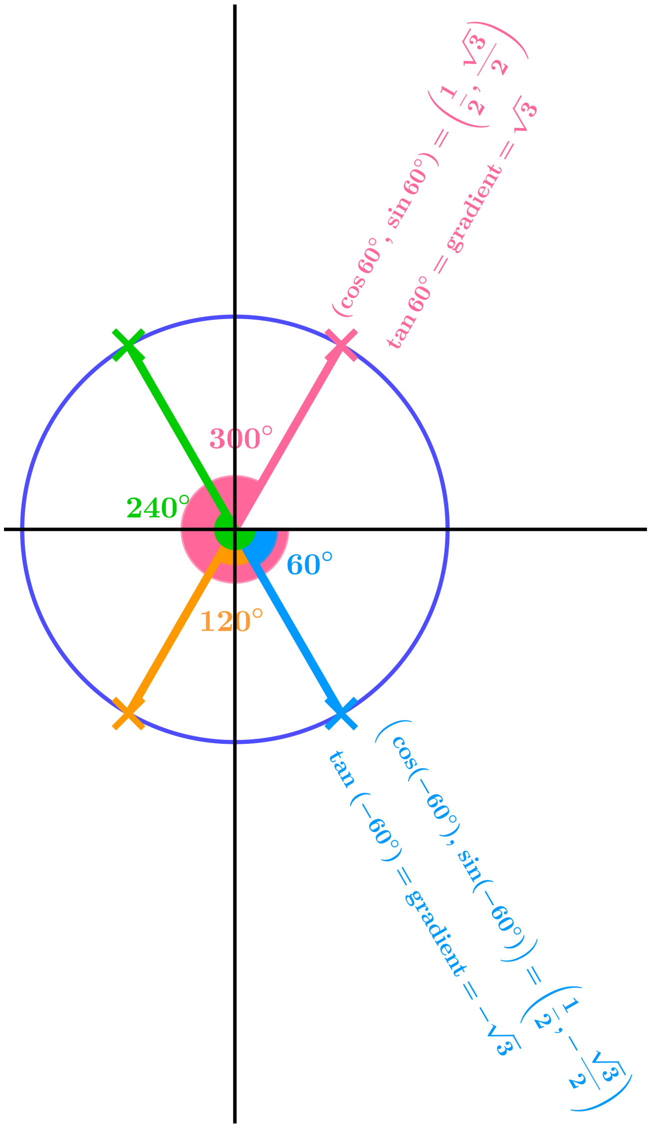


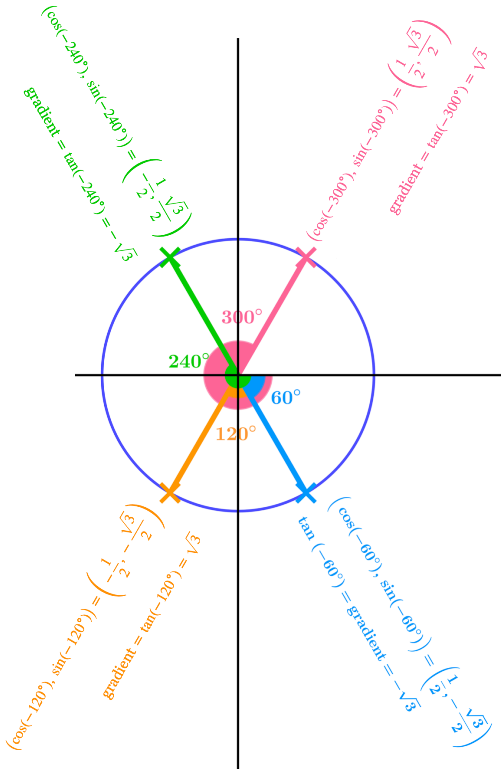
Suggest values for sin, cos, tan of $-60°$.



Suggest values for sin, cos, tan of $-120°$, $-240°$, $-300°$.

Suggest values for sin, cos, tan of $-120°$, $-240°$, $-300°$.



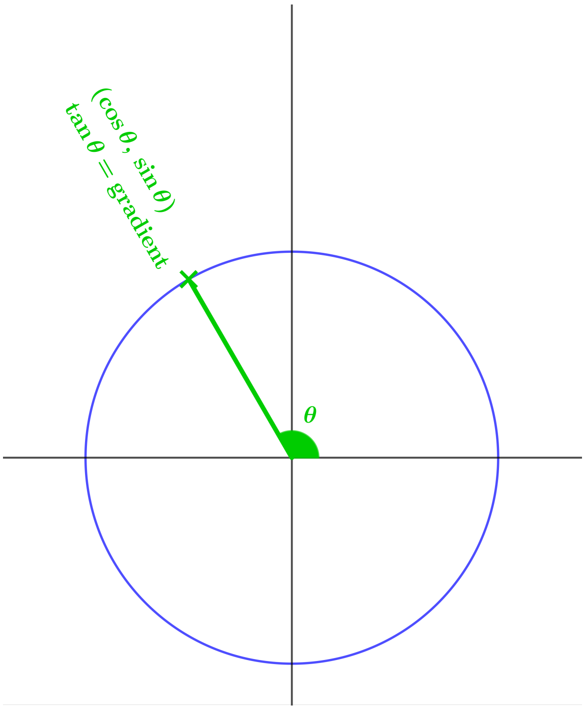


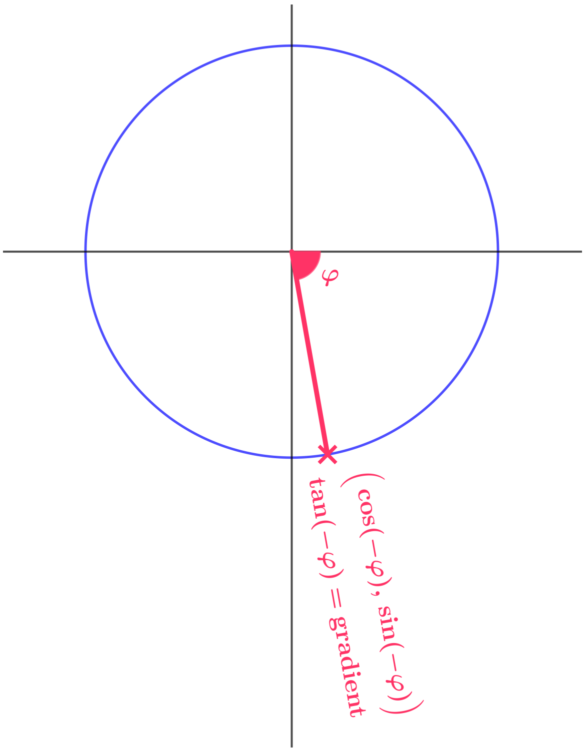
$\left(\cos(-240°), \sin(-240°)\right) = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
gradient $= \tan(-240°) = -\sqrt{3}$
$\left(\cos(-300°), \sin(-300°)\right) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
gradient $= \tan(-300°) = \sqrt{3}$
300°
240°
60°
120°
$\left(\cos(-120°), \sin(-120°)\right) = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
gradient $= \tan(-120°) = \sqrt{3}$
$\left(\cos(-60°), \sin(-60°)\right) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$
$\tan(-60°) =$ gradient $= -\sqrt{3}$

Suggest values for sin, cos, tan of $-660°$, $-1020°$, $-1380°$, $-1740°$.

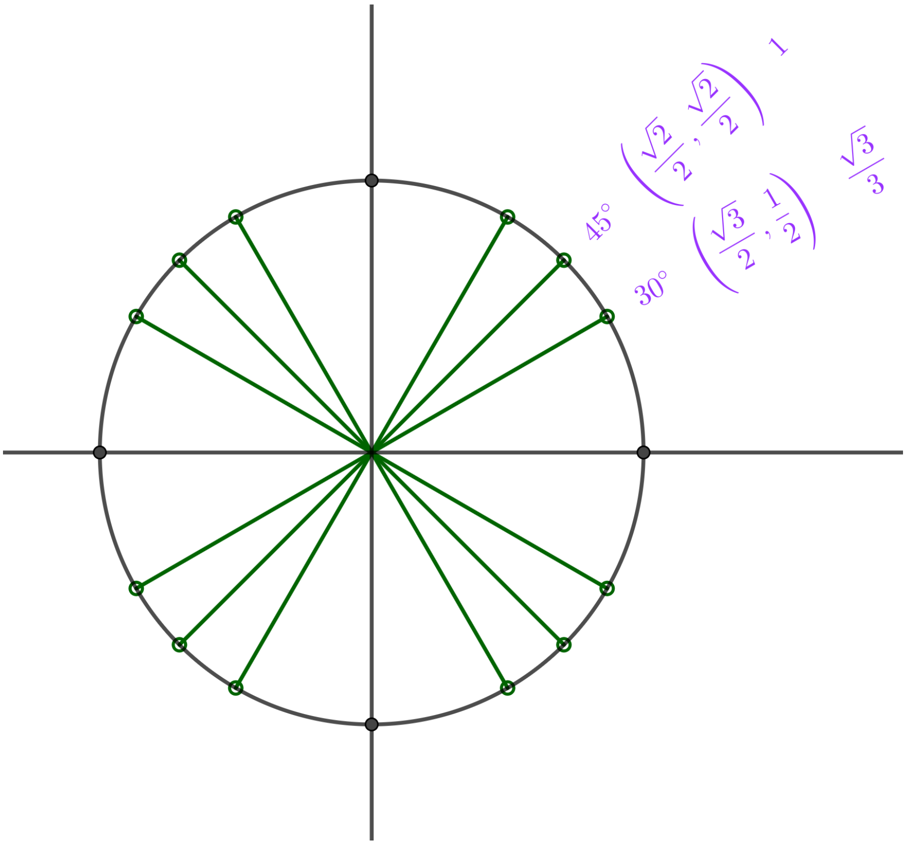
These diagrams essentially form definitions of sin, cos, and tan of any number (angle).

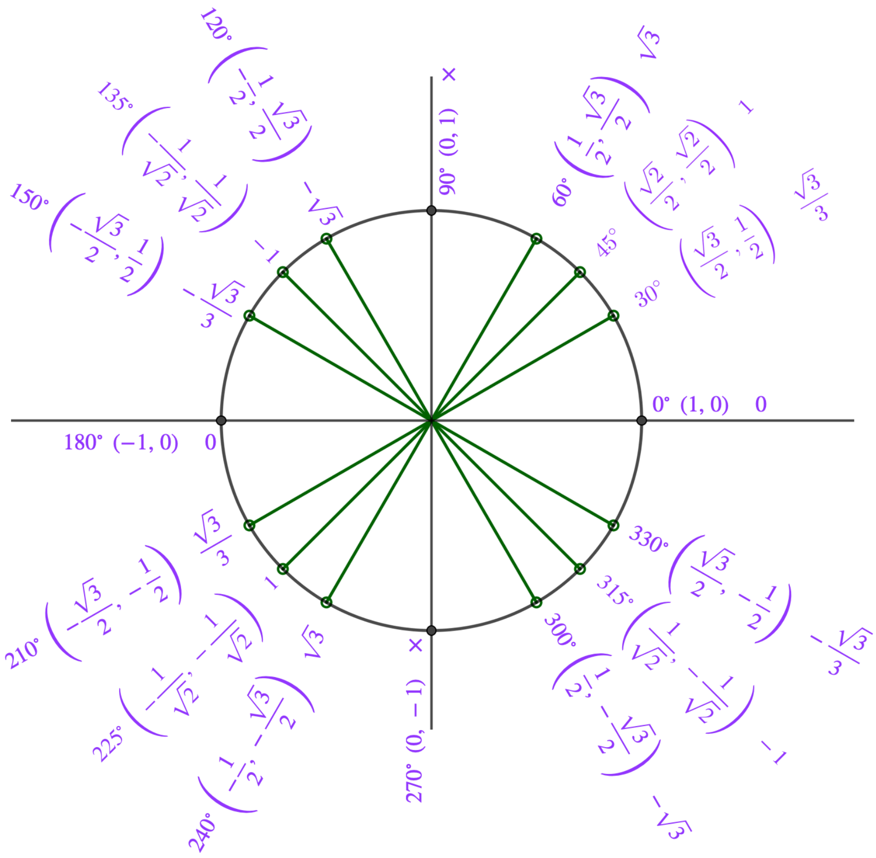
There are other ways to define these functions using, for example, graphs or power series, but this one makes intuitive sense and forms a reliable foundation for what is to come.





Complete this diagram.





My top tip: make sure you can complete a blank version of this diagram in 5 minutes or less. If you can do this, you will find all the rest of your work on this topic goes much more smoothly. And you will feel so good about not having to rely on your calculator.